UrbanGenNet: Domain-Generalized Remote Sensing Semantic Segmentation via Decoupled Style Perturbation and Class-Wise Invariant Learning

Mengyi Zhang, Wang Tian-Ge, and Zijun Zhang, *Senior Member, IEEE*

***Abstract*—Coupled radiometric and geographic shifts challenge domain-generalized semantic segmentation for optical remote sensing. The former arises from spatially nonstationary imaging conditions, such as shadows, illumination gradients, and sensor-dependent appearance changes, while the latter is reflected in city-dependent scene layouts and context priors. Existing domain generalization (DG) methods can improve appearance robustness through style/statistics diversification, but they do not explicitly model the spatially heterogeneous radiometric artifacts common in remote sensing imagery. Meanwhile, existing invariance-oriented methods do not directly enforce geographically stable semantic prediction in dense scenes, where layout and contextual priors may vary substantially across regions. To address these two limitations jointly, we propose UrbanGenNet, a two-stream remote sensing-specific framework with a shared backbone. An augmented stream applies spatially aware style perturbation (SASP) to emulate realistic spatially varying radiometric disturbances. A clean stream preserves unperturbed features for optimizing an urban loss, which performs class-wise invariant regularization across CLIP-guided latent environments to encourage geographically stable semantic prediction. By decoupling radiometric diversification from environment-based structural invariance learning, UrbanGenNet enables the backbone to acquire both robustness to nonstationary radiometric variation and invariance to geographic/contextual shift without mutual interference. Experimental results show that UrbanGenNet outperforms recent representative methods on WHU-Mix and remains highly competitive on FLAIR #1, showing complementary advantages in small-object completeness and robustness under shadow-like perturbations.**

***Index Terms*—Building extraction, domain generalization (DG), land-cover mapping, remote sensing, semantic segmentation.**

Received 8 April 2026; revised 29 May 2026 and 18 July 2026; accepted 21 July 2026. Date of publication 24 July 2026; date of current version 6 August 2026. This work was supported in part by the Smart Grid-National Science and Technology Major Project under Grant 2026ZD0809800, in part by Hong Kong Research Grants Council (RGC) General Research Fund Project under Grant 11213124, in part by the Hong Kong Research Grants Council Theme-based Research Scheme Project under Grant T43-101/26-N, in part by the Hong Kong Research Grants Council Collaborative Research Fund Project under Grant C1049-24GF, in part by Shenzhen–Hong Kong–Macau Science and Technology Category C Project under Grant SGDX20220530111205037, and in part by Shanghai Science and Technology Project under Grant 25ZR1402508. *(Corresponding authors: Wang Tian-Ge; Zijun Zhang.)*

Mengyi Zhang and Zijun Zhang are with the Department of Data Science, College of Computing, City University of Hong Kong, Hong Kong (e-mail: zijzhang@cityu.edu.hk).

Wang Tian-Ge is with the College of Transportation and the State Key Laboratory of High-Speed Maglev Transportation Technology, Tongji University, Shanghai 201804, China (e-mail: wangtiange@tongji.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3716655

I. Introduction

SEMANTIC segmentation assigns a semantic label to each pixel and is a core technique for optical remote sensing applications such as large-scale urban planning, infrastructure monitoring, and automatic building extraction from high-resolution imagery. Despite the rapid progress of deep segmentation models, robust deployment at a global scale remains challenging: models trained on a limited set of cities and imaging conditions often degrade substantially when transferred to unseen regions or sensors [1], [2]. In remote sensing, such domain shifts are typically driven by two intertwined factors. First, nonstationary radiometric variations arise from sensors, illumination, seasonal effects, atmospheric conditions, and localized occlusions (e.g., shadows), which introduce complex, spatially varying appearance changes. Second, geographic structural variations (e.g., city-dependent architectural patterns, road/building layouts, and background context priors) alter the semantic composition and spatial structure of scenes. These coupled shifts pose a major obstacle to reliable cross-city and cross-sensor semantic segmentation.

Domain adaptation (DA) and domain generalization (DG) have been widely studied to mitigate distribution shifts. However, DA typically assumes access to the target domain and requires repeated fine-tuning, which is impractical in remote sensing, where new regions and sensors continuously emerge worldwide [3], [4], [5], [6], [7]. DG, in contrast, aims to learn models that generalize to unseen target domains without target data access [8], [9], making it a more appealing and practical paradigm for remote sensing. Nevertheless, achieving reliable out-of-distribution generalization in optical remote sensing remains fundamentally challenging because the domain shift is often driven by coupled radiometric and geographic factors. Radiometric shift is induced by spatially nonstationary imaging conditions, such as illumination changes, shadows, atmospheric effects, and sensor-dependent appearance variations, whereas geographic shift is reflected in city- or region-dependent scene layouts, object arrangements, and contextual priors. Existing DG methods can improve robustness through appearance diversification, invariance regularization, or recent foundation-model adaptation. However, they are not explicitly designed to jointly address these coupled shifts, particularly the combination of spatially heterogeneous radiometric artifacts and geographically varying

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

structural/contextual priors in a unified remote sensing-specific framework.

To address the aforementioned challenges, we propose UrbanGenNet, a unified framework that tackles these issues through three coordinated components. First, for radiometric robustness, we design spatially aware style perturbation (SASP), which synthesizes spatially varying radiometric perturbations to mimic realistic appearance disturbances, such as localized shadows and exposure gradients. Second, for geographic generalization, we introduce urban loss, a task-specific objective that combines class-wise invariance learning across automatically discovered environments with boundary-aware supervision. In this way, the model is encouraged not only to reduce reliance on environment-specific correlations, but also to preserve accurate local delineation in dense remote sensing prediction. Third, to reduce mutual interference between perturbation-based robustness learning and invariance regularization, we adopt a decoupled two-stream pipeline: a clean stream without SASP optimizes urban loss for structure-oriented invariance learning, while an augmented stream with SASP focuses on radiometric diversification.

Our main contributions in this work are summarized as follows.

1) We propose UrbanGenNet, a decoupled two-stream framework for remote sensing DG semantic segmentation, designed to jointly address spatially nonstationary radiometric shifts and geographic structural variation.
2) We design the SASP module which synthesizes spatially varying radiometric perturbations (e.g., localized shadows and exposure gradients) to improve robustness to radiometric shifts.
3) We introduce urban loss, a task-specific objective that combines class-wise invariance learning with boundary-aware supervision for dense remote sensing prediction.
4) We validate UrbanGenNet on two complementary remote sensing DG benchmarks, WHU-Mix and FLAIR #1, and provide extensive quantitative, qualitative, fine-grained, and ablation analyses.

II. Related Work

A. Semantic Segmentation Under Domain Shift

Existing domain-shift segmentation methods include both target-access DA and target-free DG. Representative unsupervised DA (UDA) methods, such as DAFormer [10] and HRDA [11], provide useful References for architecture design, high-resolution context modeling, and training strategies under domain shift. Other target-access adaptation studies improve prediction stability through online self-training [12] or target-domain pseudo-relation learning [13]. However, they rely on unlabeled target-domain samples during training and are therefore different from the target-free DG setting considered in this work. In the following, we focus on DG semantic segmentation and organize existing methods into three representative lines: feature-level diversification and normalization, learning strategies and invariance regularization, and vision-foundation-model (VFM) adaptation.

1) Feature-Level Diversification and Normalization: A dominant line of DG research addresses appearance shifts by directly manipulating feature statistics or suppressing domain-specific style cues. Style augmentation methods explicitly synthesize novel domains; for instance, MixStyle [14] probabilistically interpolates the channel-wise mean and variance of feature maps across instances, while WildNet [15] injects randomized style variations. Concurrently, normalization and whitening techniques aim to filter out domain-specific appearance factors. IBN-Net [16] combines instance normalization and batch normalization to balance style invariance and discriminative representation learning, whereas RobustNet [17] and SAN-SAW [18] further regularize style-sensitive correlations through whitening-based mechanisms. These studies have established feature diversification and normalization as an important direction for improving robustness to appearance variation. For semantic segmentation, they also provide a useful foundation for studying how style-sensitive information can be modulated while maintaining discriminative spatial representations.

2) Learning Strategies and Invariance Regularization: Another line of work improves DG through training strategies and invariance-oriented learning. Meta-learning methods such as MLDG [19] simulate domain shifts by splitting source domains into meta-train and meta-test subsets, thereby encouraging models to generalize across gradient conflicts induced by domain variation. Invariant risk minimization (IRM) [20] further promotes environment-invariant prediction by explicitly constraining the optimal classifier to remain stable across environments, thus encouraging the extraction of causal rather than spurious features. This line of research complements feature-level diversification by emphasizing stable predictive mechanisms across environments. For semantic segmentation, such ideas are particularly relevant because robust generalization should depend not only on appearance robustness, but also on prediction rules that remain consistent under domain shift.

3) VFMs for DG: Recent advances in domain-generalized semantic segmentation have increasingly shifted from conventional backbones toward VFM adaptation. Rein [21] is a representative example that systematically studies frozen VFMs in DGSS and shows that parameter-efficient refinement can substantially improve generalization under a strong pretrained visual backbone. Beyond single-backbone adaptation, CLOUDS [22] further explores the collaboration of multiple foundation models, including CLIP [23], generative models, and SAM [24], to improve DGSS through robust representation, content diversification, and pseudo-label refinement. These studies demonstrate that foundation models provide a strong paradigm for DG segmentation. However, they are mainly developed for general scene segmentation and do not explicitly target the coupled radiometric and geographic shifts that characterize optical remote sensing imagery.

B. Remote Sensing Semantic Segmentation for Urban Mapping

1) Remote Sensing-Specific DG Segmentation: The above DG ideas have also been extended to remote sensing semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

segmentation. In particular, FOSMix [25] investigates domain shift in the frequency domain and introduces an adaptive frequency-based style mixing strategy for remote sensing semantic segmentation. By learning which frequency components should be preserved and which should be mixed with a reference image, it makes style diversification more flexible than fixed frequency decomposition. Multilayer DG [26] introduces progressive adversarial domain alignment via gradient reversal layers at multiple encoder stages, showing that domain consistency can be encouraged beyond bottleneck representations alone. More recently, the geospatial semantic adapter (GeoSA)-batch style augmenter (BaSA) [27] systematically investigates VFMs for cross-domain remote sensing segmentation and, based on a strong DINOv2 [28] backbone, introduces two remote sensing-specific modules: the GeoSA for feature adaptation and the BaSA for style diversification. CrossEarth [29] further extends this direction by introducing a geospatial VFM for remote sensing domain-generalizable semantic segmentation, with explicit consideration of domain gaps induced by region, sensor, wavelength, and imaging conditions.

2) Benchmarking Cross-Region Shifts in Urban Mapping: Semantic segmentation of optical remote sensing imagery is the cornerstone of large-scale urban mapping. However, models trained on isolated regions frequently fail when deployed to new areas due to structural variations in architectural layouts and nonstationary sensor conditions. To explicitly evaluate generalization against these geographic domain shifts, specialized cross-region benchmarks are strictly required. For building extraction, the WHU-Mix benchmark [2] was constructed by gathering diverse aerial images from distinct cities, explicitly testing cross-city structural robustness.

Similarly, for large-scale land-cover segmentation, models must categorize diverse semantic classes (e.g., vegetation and impervious surfaces) across varying geographic landscapes. The FLAIR #1 dataset [30] provides a large-scale country-level aerial benchmark with substantial geographic, radiometric, and temporal diversity across French spatial domains. Following prior DG practice on FLAIR #1, such as FOSMix [25], region-disjoint partitions can be used to emulate cross-region domain shift for multiclass land-cover segmentation. Therefore, WHU-Mix and FLAIR #1 serve as highly complementary, rigorous testbeds for evaluating out-of-distribution generalization.

Taken together, the above studies provide the methodological and application background for this work. Existing domain-shift segmentation methods have progressed from conventional feature diversification and invariance learning toward stronger VFM adaptation, while remote sensing semantic segmentation has been widely studied for practical urban mapping tasks such as building extraction and land-cover mapping. Compared with these works, UrbanGenNet focuses on a complementary remote sensing DG problem formulation: it aims to improve semantic prediction under coupled radiometric and geographic shifts. Specifically, the proposed framework combines spatially nonstationary radiometric diversification with geographically stable semantic learning, targeting localized radiometric artifacts such as shadows and illumination gradients, as well as region-dependent layout and contextual variation in optical remote sensing imagery.

III. Methodology

A. Motivation and Overall Architecture

Optical remote sensing DG is affected by two coupled shifts: spatially heterogeneous radiometric variation and geographically induced structural or contextual variation. The former arises from factors such as uneven illumination, seasonal appearance changes, and sensor-dependent radiometric differences, which are often locally nonstationary within a scene. This makes it desirable to explicitly model spatially varying appearance perturbation, rather than relying only on globally uniform style manipulation. UrbanGenNet therefore introduces the SASP module to enrich radiometric diversity in a spatially aware manner.

At the same time, robustness to radiometric variation alone does not ensure generalization across regions. Geographic shift is also reflected in scene layout, background composition, and context priors, which makes environment-based invariance a natural direction. However, predefined geographic metadata, such as city labels, may not faithfully characterize the latent environments that are most relevant to structural generalization: samples from different cities can share similar layouts, while substantial structural variation may also occur within the same city. For this reason, urban loss first performs metadata-free latent environment discovery. More importantly, environment discovery alone is not sufficient. Unlike image-level classification, remote sensing semantic segmentation is a dense prediction task in which class distributions and valid pixel counts can vary substantially across environments, making vanilla IRM not directly suitable for this setting. We therefore introduce a task-specific urban loss that performs class-wise invariant regularization over the discovered environments, so that environment-based consistency can be enforced in a way that is compatible with dense prediction and local structural delineation.

These two components address different aspects of the DG problem, but they are not most naturally optimized on the same feature stream. Features subjected to strong appearance perturbation are useful for improving radiometric robustness, but they may weaken the natural contextual cues needed for reliable environment-based regularization. Conversely, features that preserve the original spatial context are better suited for environment-based learning, but do not by themselves provide sufficient radiometric diversification. UrbanGenNet therefore integrates the above components within a decoupled two-stream pipeline (Fig. 1 and Algorithm 1), where an augmented stream is responsible for spatial style diversification and a clean stream preserves unperturbed spatial context and applies urban loss under the precomputed latent environments. In this way, SASP, metadata-free environment discovery, and urban loss play complementary roles, while the two-stream pipeline serves as the training mechanism that allows these objectives to be optimized jointly in a unified framework.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

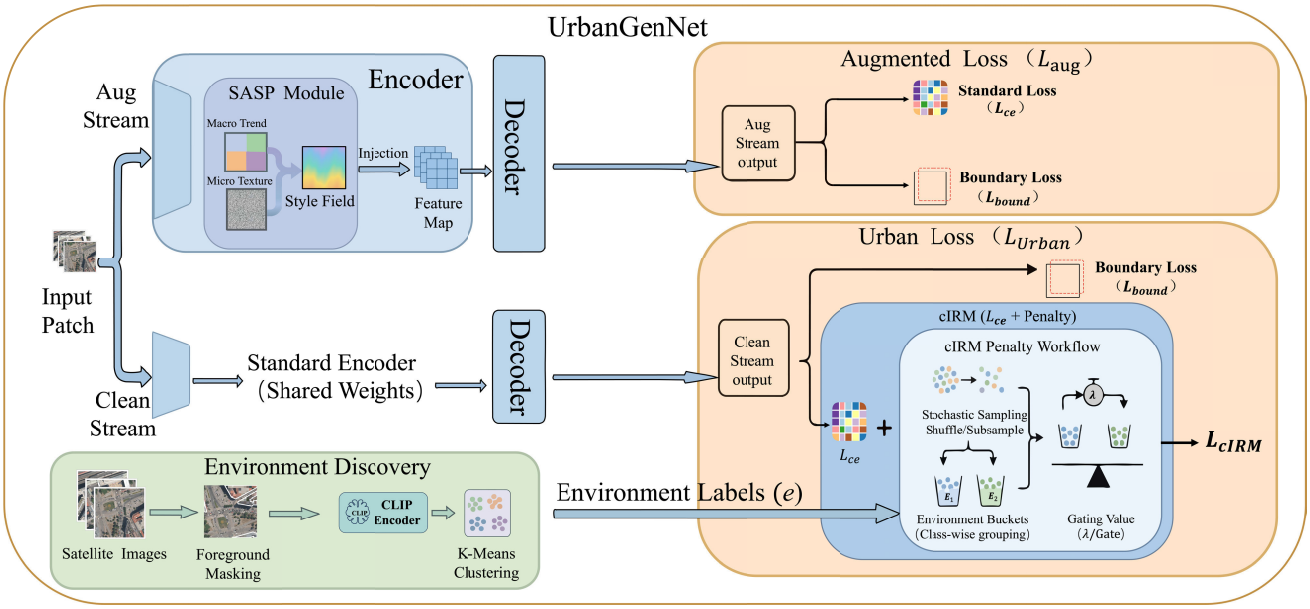


Fig. 1. Overall architecture of the proposed UrbanGenNet. The shared backbone bifurcates into an augmented stream with the SASP module for radiometric robustness and a clean stream utilizing latent environment discovery without geographic metadata and urban loss for causal geographic invariance.

B. Augmented Stream: Spatially Aware Style Perturbation

Real-world radiometric variations in remote sensing exhibit spatial structure: shadows cast by clouds create smooth gradients, while sensor artifacts introduce localized patterns. To capture this, SASP decomposes style perturbations into two components: 1) a smooth macro-level spatial trend that models large-scale variations (e.g., illumination gradients); and 2) fine-grained micro-level noise that introduces local textural diversity, which reduces the model's reliance on source-specific local radiometric correlations that are predictive in the training domains but unstable under cross-domain shift. SASP generates spatially varying radiometric perturbations that better match real remote sensing artifacts than global style mixing.

Given an intermediate feature map $F \in \mathbb{R}^{B\times C\times H\times W}$, we hypothesize that a realistic radiometric shift can be decomposed into two components: a *macro-level spatial trend* and *micro-level textural variations*.

We partition the spatial dimensions into a coarse $G \times G$ grid and compute local statistics $\mu_{grid}, \sigma_{grid}$ for each cell from permuted batch samples. These discrete grids are then bilinearly upsampled to form continuous macro-trend maps $\mu_{trend}, \sigma_{trend} \in \mathbb{R}^{B\times C\times H\times W}$, thereby producing a smooth low-frequency radiometric field that mimics structured illumination changes commonly observed in remote sensing imagery.

As demonstrated in our subsequent ablation studies, setting $G = 2$ provides the optimal balance between spatial variation and feature integrity.

To inject micro-level textural variations, we sample low-frequency noise grids $z_\mu, z_\sigma \in \mathbb{R}^{B\times C\times k\times k}$ (with a small grid size k, e.g., 16) from a standard Gaussian distribution $\mathcal{N}(0, \mathbf{I})$. To ensure the spatial smoothness characteristic of natural optical artifacts, these grids are upsampled via bicubic interpolation to match the target spatial dimensions, yielding smooth spatial noise fields $\tilde{z}_\mu, \tilde{z}_\sigma \in \mathbb{R}^{B\times C\times H\times W}$

$$\tilde{z}_\mu = \mathrm{UP}_{\mathrm{bicubic}}\left(z_\mu\right), \quad \tilde{z}_\sigma = \mathrm{UP}_{\mathrm{bicubic}}\left(z_\sigma\right). \tag{1}$$

So that local textural variation is introduced in a spatially coherent manner rather than as independent pixel-wise noise. The final spatially nonstationary style maps are synthesized by modulating the macro trends with the micro noise

$$\begin{aligned} \mu_{spatial} &= \mu_{trend} + \left(\tilde{z}_\mu \odot \sigma_{trend} \cdot \epsilon\right) \\ \sigma_{spatial} &= \sigma_{trend} \odot \left(1 + \tilde{z}_\sigma \cdot \epsilon\right) \end{aligned} \tag{2}$$

where $\odot$ denotes element-wise multiplication, and ϵ controls the noise intensity.

Finally, a mixing weight $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ interpolates between the original global statistics of F and the synthesized spatial style maps. The perturbed feature F_{pert} is obtained via spatial adaptive instance normalization

$$\begin{aligned} F_{pert} = &\left(\frac{F - \mu(F)}{\sigma(F)}\right) \odot \left[\lambda\sigma(F) + (1-\lambda)\,\sigma_{spatial}\right] \\ &+ \left[\lambda\mu(F) + (1-\lambda)\,\mu_{spatial}\right]. \end{aligned} \tag{3}$$

To further clarify the effect of SASP, we provide a feature-level visualization in Fig. 2. Specifically, we extract the intermediate feature before SASP and visualize how the proposed grid-based style perturbation changes the local feature-style statistics. The 2×2 style grid is constructed from training-set reference images with different radiometric appearances. SASP then interpolates these grid-level statistics into a spatially varying style field and applies it to the feature map. The resulting feature-style difference is computed as

$$\begin{aligned} D(h,w) = &\left\|\mu_{\mathrm{loc}}\left(F_{\mathrm{SASP}}\right) - \mu_{\mathrm{loc}}(F)\right\|_2 \\ &+ \beta\left\|\sigma_{\mathrm{loc}}\left(F_{\mathrm{SASP}}\right) - \sigma_{\mathrm{loc}}(F)\right\|_2 \end{aligned} \tag{4}$$

where F and F_{SASP} denote the clean and SASP-perturbed features, respectively, $\mu_{\mathrm{loc}}(\cdot)$ and $\sigma_{\mathrm{loc}}(\cdot)$ denote the local mean

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Decoupled Two-Stream Training of UrbanGenNet

Require: Training dataset $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^{N}$, pretrained CLIP encoder E_{CLIP}, shared network $f_\theta = D \circ E$, environments K, grid size G.

Ensure: Optimized network parameters θ.

// Step 1: Metadata-Free Latent Environment Discovery (Offline)
for each $(X_i, Y_i) \in \mathcal{D}$ **do**
  Extract semantic-free embedding: $v_i = E_{\text{CLIP}}(X_i \odot M_{bg})$.
end for
Cluster $\{v_1, \ldots, v_N\}$ into K environments $\mathcal{E}$ via K-means.
// Step 2: Decoupled Mini-Batch Optimization
for each training iteration **do**
  Sample mini-batch (X, Y) and their environments $e \in \mathcal{E}$.
  # 1. Shallow Feature Extraction
  Extract intermediate features: $F_{mid} = E_1(X)$
  # 2. Clean Stream (Causal Structure & Boundary Preservation)
  Forward through deep encoder: $F_{clean} = E_2(F_{mid})$
  Clean predictions: $\Phi_{clean} = D(F_{clean})$
  Compute base empirical loss and Boundary IoU: $\mathcal{L}_{CE}^{clean}, \mathcal{L}_{BIoU}^{clean}$
  Compute class-wise gradient penalty $\mathcal{L}_{env}$ using partitioned subsets (Eq. 6-7).
  Formulate clean objective: $\mathcal{L}_{Urban} = \mathcal{L}_{CE}^{clean} + \lambda_b \mathcal{L}_{BIoU}^{clean} + \lambda_{cIRM} \mathcal{L}_{env}$
  # 3. Augmented Stream (Spatially-Aware Diversification & Boundary Preservation)
  Extract $G \times G$ macro-level spatial trends from permuted F.
  Inject micro-level textural noise to synthesize style maps (Eq. 3).
  Apply Spatial Perturbation: $\tilde{F}_{mid} = \text{SASP}(F_{mid})$
  Forward through deep encoder: $F_{pert} = E_2(\tilde{F}_{mid})$
  Perturbed predictions: $\Phi_{pert} = D(F_{pert})$
  Formulate perturbed objective: $\mathcal{L}_{aug} = \mathcal{L}_{CE}^{aug} + \lambda_b \mathcal{L}_{BIoU}^{aug}$
  # Shared Optimization
  Compute Total Objective: $\mathcal{L}_{total} = \frac{1}{2}\mathcal{L}_{Urban} + \frac{1}{2}\mathcal{L}_{aug}$
  Update θ via backpropagation on $\mathcal{L}_{total}$.
end for

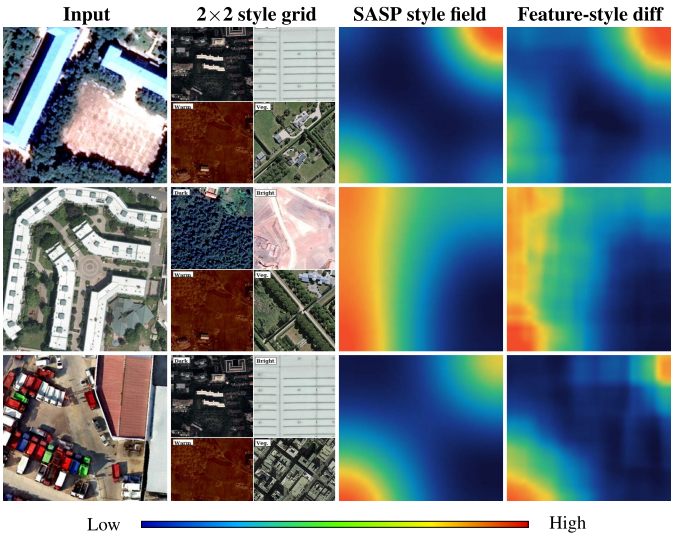


Fig. 2. Visualization of SASP-induced feature-style perturbation on WHU-Mix training samples.

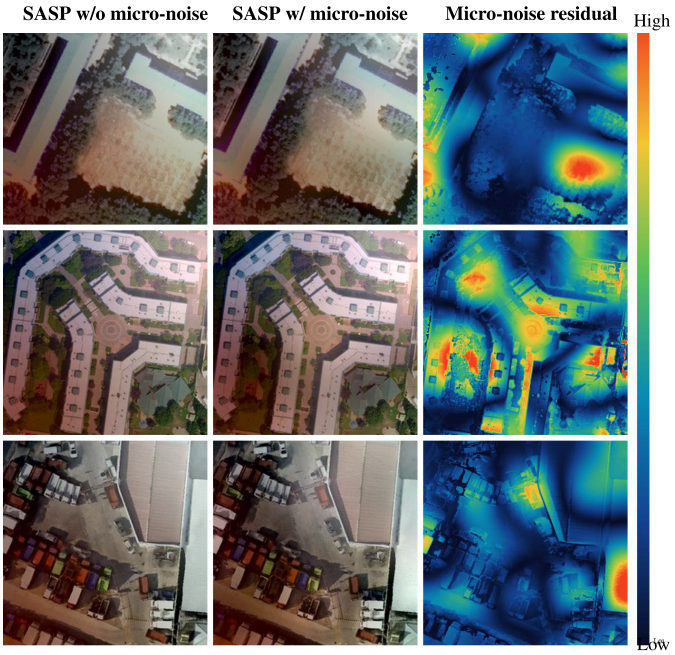


Fig. 3. Visualization of the micro-noise effect in SASP on WHU-Mix training samples.

and standard deviation, and β controls the contribution of the local standard-deviation difference.

As shown in Fig. 2, the resulting feature-style difference is spatially nonuniform rather than globally homogeneous. Stronger responses appear in different local regions according to the imposed spatial style field. This behavior is consistent with the motivation of modeling spatially nonstationary radiometric variation in high-resolution remote sensing imagery.

We further visualize the effect of the micro-noise component in Fig. 3. Compared with SASP without micro-noise, adding micro-noise introduces mild local appearance variations while preserving the overall image structure. The residual maps show spatially nonuniform but smooth responses, rather than independent pixel-level noise. This observation is consistent with the design of micro-noise in SASP, which provides low-frequency local radiometric variation on top of the grid-based spatial style field.

C. Clean Stream: Metadata-Free Latent Environment Discovery

To perform environment-based invariant learning in the clean stream, the training data must first be partitioned into environments that reflect meaningful contextual variation. Using coarse geographic metadata such as city labels is not ideal for this purpose: samples from different cities may still share similar scene layouts or background patterns, while substantial structural variation can also exist within the same city. For this reason, we seek a metadata-free environment partition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

that is guided by visual context rather than administrative identity.

We employ a frozen CLIP image encoder E_{CLIP} to extract contextual embeddings for latent environment discovery. Our choice of CLIP is motivated by its strong transferability and semantically rich visual representations learned from large-scale vision–language pretraining, which make it a suitable generic encoder for context-level clustering. Importantly, our goal here is not to interpret CLIP features as explicit geographic labels, but to use them as a robust embedding space in which samples with similar contextual characteristics can be grouped more naturally.

Crucially, to ensure that the clustering process is driven strictly by these underlying environmental signatures rather than being biased by task-specific foreground objects (e.g., specific building roof textures or colors causing semantic leakage), we apply a background mask M_{bg} derived from the ground-truth labels. Specifically, we define the background mask M_{bg} by excluding all foreground semantic classes (e.g., buildings, roads, and vehicles) and retaining only the contextual elements (e.g., vegetation, terrain, and sky). This ensures that the clustering is driven by geographic context rather than object-specific features. The semantic-free contextual embedding for image X_i is explicitly extracted as $v_i = E_{\text{CLIP}}(X_i \odot M_{bg})$. We then apply K-means to cluster $\{v_1, \ldots, v_N\}$ into K latent environments $\mathcal{E} = \{e_1, \ldots, e_K\}$. The number of environments K is treated as a prior hyperparameter reflecting the underlying geographic diversity of the source domains.

D. Clean Stream: Class-Wise IRM

Given the discovered latent environments, the remaining issue is how to impose cross-environment consistency in a way that is compatible with dense prediction. In this work, we instantiate environment-based regularization using IRM [20]. However, directly applying vanilla IRM to semantic segmentation is unsatisfactory. Unlike image-level classification, dense prediction is performed at the pixel level, where the number of valid pixels can vary substantially across classes and environments. As a result, a single environment-level IRM penalty may be dominated by large class–environment buckets, while extremely sparse buckets can produce highly noisy and unstable gradients.

To address this issue, we formulate a class-wise IRM (cIRM) objective tailored to semantic segmentation. Instead of computing one invariance penalty over all pixels jointly, cIRM constructs IRM penalties within each class–environment bucket and stabilizes their estimation before cross-environment aggregation.

For a class $c \in \mathcal{C}$ and an environment $e \in \mathcal{E}$, let Ω_c^e be the set of valid pixels belonging to class c in environment e, and let $N_c^e = |\Omega_c^e|$ be the corresponding pixel count. Since extremely large buckets incur unnecessary computation and can dominate the penalty estimate, while extremely small buckets lead to high-variance estimates, we first control the effective sample size at the bucket level. Concretely, if $N_c^e > N_{\max}$, we randomly subsample $N_{\max}$ pixels from Ω_c^e; otherwise all available pixels are retained. The selected pixels are then randomly partitioned into two disjoint subsets $S_1^{c,e}$ and $S_2^{c,e}$ of equal size. This random split provides two stochastic views of the same class–environment bucket and avoids the spatial bias that may arise from deterministic partitioning.

Following the IRMv1 paradigm [20], we introduce a dummy scalar classifier $w \equiv 1.0$ and use it only as a probe variable. Let Φ_c be the logits associated with class c, and let $\mathcal{R}_1^{c,e}$ and $\mathcal{R}_2^{c,e}$ be the empirical cross-entropy risks computed on $S_1^{c,e}$ and $S_2^{c,e}$, respectively. The basic class-wise IRM penalty is defined as

$$P_{c,e} = \nabla_w \mathcal{R}_1^{c,e}(w \cdot \Phi_c) \cdot \nabla_w \mathcal{R}_2^{c,e}(w \cdot \Phi_c). \tag{5}$$

This quantity serves as a stochastic IRMv1-style estimate at the class–environment level.

To make this estimate more reliable in dense prediction, we further introduce a stabilized penalty. First, under-supported buckets are softly downweighted through

$$g_{c,e} = \min\left(1, \frac{N_c^e}{N_{\min}}\right) \tag{6}$$

so that sparse buckets are not discarded entirely, but their contribution is reduced according to their effective sample support. Second, we apply a lower bound clamping operation to suppress noisy negative spikes that may arise from rare or difficult buckets. The stabilized class-wise penalty is therefore written as

$$\tilde{P}_{c,e} = g_{c,e}\,\text{clamp}\left(P_{c,e}, 0, +\infty\right). \tag{7}$$

After bucket-wise stabilization, the penalties are aggregated across environments. Since the discovered environments within a mini-batch can be highly unbalanced, directly averaging over environments would allow large environments to dominate the regularizer. We therefore introduce an environment soft-balancing term

$$\omega_e = \frac{1}{(B_e + \epsilon)^{\alpha}} \tag{8}$$

where B_e denotes the number of samples in environment e, α controls the degree of reweighting, and ϵ is a small constant for numerical stability. The final environment regularization is defined as

$$\mathcal{L}_{\text{env}} = \frac{1}{\sum_{e\in\mathcal{E}} \omega_e} \sum_{e\in\mathcal{E}} \omega_e \sum_{c\in\mathcal{C}} \tilde{P}_{c,e}. \tag{9}$$

In this formulation, the bucket-level operations in (5)–(7) are introduced to stabilize penalty estimation, whereas (8) is used to reduce aggregation bias caused by environment-size imbalance. These two roles are complementary rather than redundant.

Finally, the class-wise IRM objective combines the standard segmentation loss with the proposed cIRM regularization

$$\mathcal{L}_{cIRM} = \mathcal{L}_{CE}^{clean} + \lambda_{cIRM}\,\mathcal{L}_{env} \tag{10}$$

where λ_{cIRM} is the penalty weight. In summary, cIRM differs from vanilla IRM in three aspects that are essential for dense prediction: it computes invariance penalties at the class–environment level rather than over all pixels jointly, stabilizes each bucket-wise penalty through sample control and soft downweighting of unreliable buckets, and aggregates these penalties with environment-aware reweighting to reduce imbalance across environments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

E. Joint Optimization and Urban Loss

To preserve geometric fidelity and delineate sharp urban structures (e.g., building boundaries and road networks) under severe domain shifts, we incorporate the boundary IoU loss $\mathcal{L}_{BIoU}$ [31]. Unlike standard IoU that treats all pixels equally, boundary IoU specifically focuses on pixels near class boundaries, explicitly penalizing structural distortions along spatial contours. This is particularly important in urban remote sensing where precise boundary delineation is critical for applications such as building footprint extraction.

The decoupled two-stream architecture is optimized jointly, with each stream minimizing a distinct functional objective.

The augmented stream focuses on empirical robustness against nonstationary radiometric variations. Its objective $\mathcal{L}_{pert}$ combines the standard dense cross-entropy ($\mathcal{L}_{CE}$) and the boundary IoU loss

$$\mathcal{L}_{aug} = \mathcal{L}_{CE}\left(\Phi_{pert}, Y\right) + \lambda_b \mathcal{L}_{BIoU}\left(\Phi_{pert}, Y\right) \tag{11}$$

where λ_b is the balancing weight for the boundary penalty.

Conversely, the clean stream is designed for environment-consistent structural learning under preserved spatial context. To this end, it first performs metadata-free latent environment discovery, and then imposes a customized class-wise invariant regularization term, cIRM, over the discovered environments. Since dense prediction also requires accurate geometric delineation, we further combine cIRM with boundary-aware supervision to construct the proposed urban loss ($\mathcal{L}_{\text{Urban}}$). The clean stream is therefore supervised by $\mathcal{L}_{\text{Urban}}$, so that cross-environment regularization and boundary preservation can be optimized in a coordinated manner

$$\mathcal{L}_{Urban} = \mathcal{L}_{cIRM} + \lambda_b \mathcal{L}_{BIoU}\left(\Phi_{clean}, Y\right). \tag{12}$$

Finally, the unified total objective for updating the shared network parameters θ is formulated as the average of both streams. This decoupled two-stream pipeline ensures that the backbone simultaneously acquires spatially aware radiometric diversity from the augmented stream and strict causal geographic invariance from the clean stream

$$\mathcal{L}_{total} = \frac{1}{2}\mathcal{L}_{Urban} + \frac{1}{2}\mathcal{L}_{aug}. \tag{13}$$

IV. Experiments and Analysis

In this section, we evaluate the proposed UrbanGenNet on two challenging optical remote sensing DG tasks, namely, cross-city building extraction and large-scale land-cover mapping. We first introduce the datasets, implementation details, and evaluation protocol. We then report benchmark comparisons with recent representative methods, including foundation-model fine-tuning baselines. To further examine the behavior of recent fine-tuning-based methods in remote sensing DG, we conduct additional analyses, including fine-grained evaluation under remote sensing-specific challenging cases and a hybrid validation with a recent VFM fine-tuning pipeline. Finally, we provide extensive ablation studies to validate the effectiveness of the proposed design.

A. Datasets

To assess the generalization capability under severe geographic structural variations and spatially heterogeneous radiometric shifts, we adopt two representative benchmarks in which the training and testing samples are geographically disjoint.

1) Cross-City Building Extraction (WHU-Mix): WHU-Mix [2] is a large-scale benchmark specifically designed for out-of-distribution building extraction. It contains 43 727 training/validation images and 8402 test images, each with a spatial size of 512 × 512 pixels. To ensure strict domain shifts, the training set is collected from multiple regions and sensors, whereas the test set consists of five geographically distinct target cities across different continents, namely, US_Kitsap, CN_Wuxi, NZ_Dunedin, DE_Potsdam, and SD_Khartoum. The protocol strictly forbids geographic overlap between source and target domains, making it a challenging benchmark for evaluating both structural and radiometric generalization.

2) Large-Scale Land-Cover Mapping (Flair #1): To evaluate multiclass generalization in a more complex dense prediction setting, we further adopt the FLAIR #1 dataset [30], a country-scale aerial image dataset that covers diverse landscapes, topographies, and seasons across France. Although the original benchmark provides 18 fine-grained semantic categories, following the established protocol in [25], we use the 12 macro-classes setting in our experiments.

To avoid spatial leakage, we do not use the official random split. Instead, following the geographically disjoint evaluation protocol introduced in FOSMix [25], we reorganize the dataset according to its spatial metadata. Specifically, the training and validation sets are formed from designated source regions, while the test set is composed of entirely unseen target regions. This setting provides a more realistic evaluation of DG under intertwined structural and radiometric shifts.

B. Implementation Details and Evaluation Protocol

UrbanGenNet is implemented in PyTorch based on the MMSegmentation framework. Unless otherwise specified, we adopt ConvNeXt-Base [35], initialized from the ImageNet-21K pretrained checkpoint provided by OpenMMLab, as the default shared backbone, together with a UPerNet decoder. Since SASP, urban loss, and the two-stream pipeline are only used during training, UrbanGenNet reduces to the same single-stream ConvNeXt-B architecture as the baseline during inference. The CLIP-based environment discovery is conducted once before training as an offline preprocessing step and is not performed during iterative training or inference. Using the standard MMEngine/MMSegmentation complexity analysis protocol under an input size of 512 × 512, the deployed UrbanGenNet reports the same parameter count and FLOPs as the baseline, namely, 121-M parameters and 0.291-T FLOPs, indicating that the proposed method introduces no additional inference-time computational overhead.

During training, we employ the proposed decoupled two-stream strategy. The augmented stream activates the SASP module to synthesize spatially nonstationary radiometric variation, whereas the clean stream is kept free of perturbations and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE RESULTS OF DIFFERENT DG METHODS ON THE WHU-MIX BUILDING EXTRACTION BENCHMARK. THE BEST RESULT IS HIGHLIGHTED IN BOLD

Method	mIoU (%)	IoU (%)				
		US_Kitsap	CN_Wuxi	NZ_Dunedin	DE_Potsdam	SD_Khartoum
MixStyle [14]	60.86	76.45	80.32	84.91	25.17	40.35
IBN-Net [16]	59.91	75.77	79.29	85.96	21.13	41.97
MMLD [32]	56.77	68.06	74.10	82.02	22.64	39.94
LDDG [33]	58.02	71.21	79.92	81.25	26.53	31.37
RobustNet [17]	67.47	81.16	84.92	88.98	47.48	33.30
BSM [2]	72.06	80.93	84.04	88.62	58.16	48.94
FOSMix [25]	72.94	79.76	82.91	87.37	50.02	48.23
Rein [21]	73.88	82.39	84.23	87.97	60.56	54.24
UrbanGenNet	**76.10**	**83.35**	**85.21**	**89.57**	**66.07**	**54.61**

Rein adopts a ConvNeXt-L backbone, whereas UrbanGenNet uses a ConvNeXt-B backbone.

is used to compute the urban loss $\mathcal{L}_{\text{Urban}}$. At inference time, the network collapses into a standard single-stream forward pass and introduces no additional computational overhead.

We optimize the network using AdamW with an initial learning rate of 1×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 0.05. A layer-wise learning rate decay ratio of 0.9 is adopted. The learning rate is linearly warmed up for the first 3000 iterations and then decayed using a polynomial schedule with power 1.0 for a total of 80 000 iterations. Automatic mixed precision (AMP) is used during training. Standard data augmentations, including random resizing, random cropping, horizontal flipping, and photometric distortion, are applied uniformly. Unless otherwise specified, SASP uses a 2×2 macro-style grid ($G = 2$), a 16×16 micro-noise grid ($k = 16$), a micro-noise intensity of $\epsilon = 0.3$, and a symmetric style-mixing distribution $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\alpha = 0.5$. These settings are used throughout the main experiments.

For broad benchmarking, we follow common practice and report per-class IoU and mean IoU (mIoU) for all compared methods. In addition, to better characterize boundary quality, object completeness, and small-structure preservation under domain shifts, we further report boundary IoU, object-level F1, and small-object-oriented statistics, including small-object recall, miss rate, and mean best IoU. Since these fine-grained analyses serve as diagnostic evaluations beyond holistic benchmark scores, they are conducted on representative strong baselines rather than exhaustively on all earlier methods.

C. Comparison With Recent Representative Methods

1) Cross-City Building Extraction on WHU-Mix: We first compare UrbanGenNet with representative DG methods on WHU-Mix, including both conventional DG approaches and recent strong baselines such as FOSMix and rein. The quantitative results are summarized in Table I.

Overall, UrbanGenNet achieves the best mIoU among the methods listed in Table I, demonstrating strong cross-city generalization capability on this benchmark. In particular, compared with rein, which adopts a higher capacity ConvNeXt-L backbone, UrbanGenNet with a ConvNeXt-B backbone still attains stronger overall performance. This observation suggests that the proposed framework remains effective even when compared with recent strong pretrained-backbone baselines.

Compared with earlier style-diversification methods such as FOSMix, our framework does not rely solely on frequency- or statistics-level appearance mixing. Instead, UrbanGenNet explicitly models spatially varying radiometric perturbations and combines them with class-wise invariance learning, which is particularly relevant to cross-city remote sensing imagery where local shadows, illumination gradients, and nonstationary appearance shifts frequently co-occur.

The gains are especially notable on the more challenging target domains, such as DE_Potsdam and SD_Khartoum, where severe layout variation and illumination shifts pose substantial difficulties for cross-DG. These results indicate that the proposed decoupled design improves robustness not only in overall benchmark performance but also in difficult urban scenarios with pronounced structural and radiometric shifts.

2) Large-Scale Land-Cover Mapping on FLAIR #1: To further demonstrate the versatility of the proposed framework beyond binary building extraction, we evaluate UrbanGenNet on the FLAIR #1 geographic generalization split. The quantitative results are reported in Table II.

UrbanGenNet achieves strong and competitive performance on this large-scale multiclass benchmark, substantially outperforming conventional DG baselines and ranking among the strongest methods overall. Although rein obtains the highest overall mIoU, UrbanGenNet achieves the best results on several categories, including *water*, *deciduous*, and *vineyard*, and remains highly competitive on *building* and *impervious surface*.

A closer examination suggests that these gains are not random. The categories where UrbanGenNet is most effective are often associated with stronger spatial heterogeneity, richer local texture variation, or more pronounced radiometric shifts across geographic regions. This observation is consistent with the design motivation of our framework: by explicitly modeling spatially varying appearance perturbations and disentangling them from structural invariance learning, UrbanGenNet can better preserve semantically meaningful local patterns under geographically disjoint conditions. Therefore, the proposed framework is not limited to cross-city

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON ON THE FLAIR #1 GEOGRAPHIC GENERALIZATION SPLIT. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	mIoU	Build.	Perv. Surf.	Imperv. Surf.	Bare Soil	Water	Conif.	Decid.	Brushwood	Vineyard	Herb. Veget.	Agric. Land	Plowed Land
RobustNet	46.54	73.20	33.40	65.10	26.80	61.30	15.60	54.70	18.70	74.20	44.80	46.60	44.10
SiamDoGe [34]	46.58	73.50	34.20	64.60	28.10	62.10	13.40	54.20	19.20	74.80	45.60	44.10	45.20
FOSMix	48.51	74.00	35.40	66.20	35.20	60.90	12.70	57.40	19.60	75.70	45.20	46.00	53.60
Layer-by-Layer DG	53.10	64.50	**56.90**	66.00	35.90	52.07	**43.40**	67.20	**52.40**	38.30	**59.70**	**48.60**	51.70
Rein	**57.03**	**82.52**	49.82	**74.05**	**50.99**	76.22	25.38	64.06	35.47	67.94	53.55	48.30	**56.00**
UrbanGenNet (ours)	55.97	82.36	52.34	73.57	42.12	**77.60**	27.92	**67.75**	33.00	**75.94**	48.39	42.78	47.91

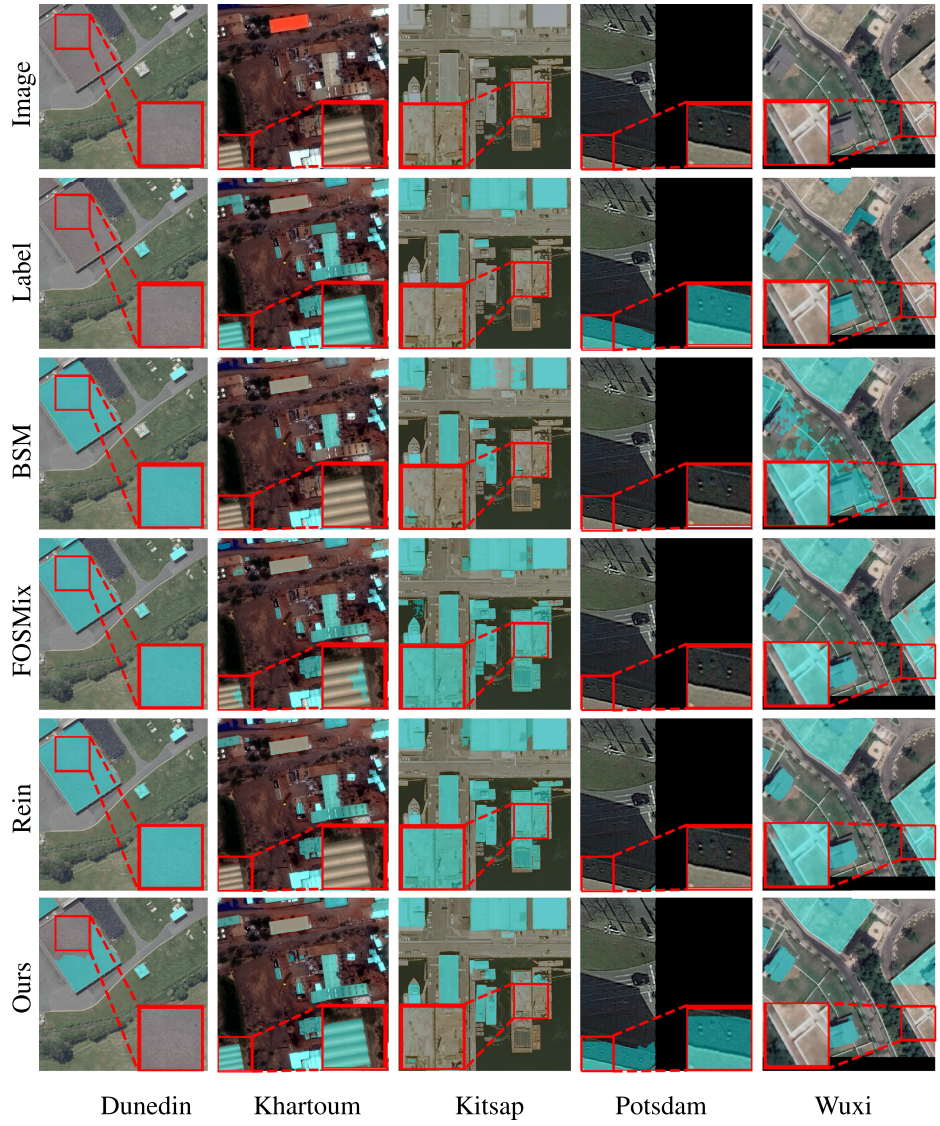


Fig. 4. Qualitative comparison on WHU-Mix. Representative examples from the five target cities are shown, together with zoomed-in views of challenging local regions. The selected cases illustrate typical failure modes under cross-city domain shifts, including small-building omission, fragmented prediction under severe illumination variation, and incomplete delineation in dense urban layouts.

building extraction, but also generalizes favorably to dense multiclass land-cover mapping with more complex semantic compositions.

3) Qualitative Comparison: To visually compare the behavior of different methods, Fig. 4 presents representative examples from the five target cities in WHU-Mix, together with zoomed-in views of challenging local regions. The selected cases cover representative local failure patterns under cross-city domain shift, including overprediction in structurally confusing regions, incomplete recovery under weak

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

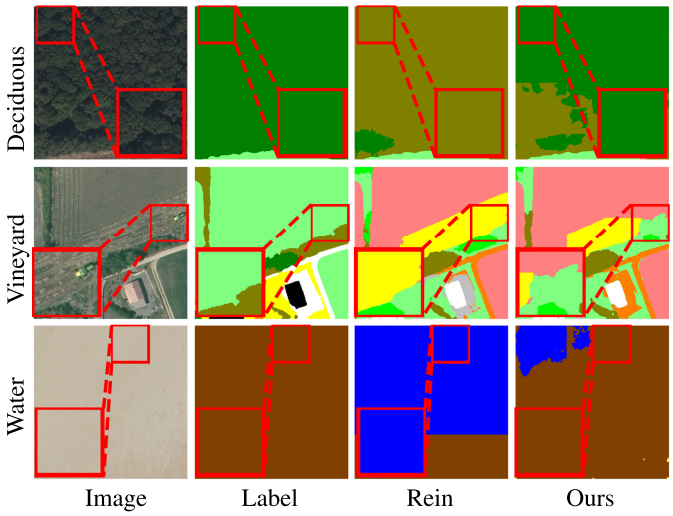


Fig. 5. Qualitative comparison on the FLAIR #1 geographic generalization split. Representative examples are shown for *deciduous*, *vineyard*, and *water*, together with highlighted local regions. Compared with rein, UrbanGenNet better preserves locally coherent semantic regions and category-specific spatial patterns under geographically disjoint conditions.

visibility or severe shadow, and imprecise delineation in complex urban scenes.

Overall, UrbanGenNet shows clearer advantages in these challenging cases by producing more spatially coherent masks and more accurate local delineation. Compared with the baseline methods, it better suppresses overprediction in confusing local layouts while preserving more complete building responses in visually difficult regions. These visual observations are consistent with the quantitative results in Table I.

To further illustrate the category-wise behavior on FLAIR #1, Fig. 5 presents representative examples from three categories where UrbanGenNet shows clear advantages, namely, *deciduous*, *vineyard*, and *water*. Compared with rein, UrbanGenNet produces more locally coherent semantic regions and better preserves category-specific spatial patterns under geographically disjoint conditions.

These visual differences are particularly evident in the challenging local regions highlighted by the red boxes. For *deciduous*, UrbanGenNet yields a more spatially coherent prediction with reduced category confusion. For *vineyard*, it improves local prediction quality in the highlighted parcel region and around structural boundaries. For *water*, it suppresses the large spurious response observed in rein and produces a result closer to the ground truth. Overall, these examples suggest that the category-wise gains of UrbanGenNet are related to the roles of its two main components. For categories such as *water*, which are highly sensitive to spatially varying radiometric disturbances, the improvement is consistent with the radiometric robustness introduced by SASP. For *deciduous* and *vineyard*, where prediction quality depends more strongly on geographically varying contextual priors and locally structured semantic patterns, the gains are more consistent with the geographically stable semantic learning encouraged by urban loss. These observations are also consistent with the quantitative results in Table II.

D. Fine-Grained Analysis Against Foundation-Model Fine-Tuning Methods

To further analyze the behavior of recent foundation-model fine-tuning methods in remote sensing DG, we conduct two complementary evaluations against representative strong baselines. First, we compare UrbanGenNet with GeoSA-BaSA using object- and small-structure-oriented metrics under clean and shadow-like perturbation settings, in order to examine performance on remote sensing-specific challenging cases. Second, we perform a hybrid validation and additional fine-grained verification based on rein under a unified DINOv2-based protocol to evaluate whether the proposed modules remain effective when integrated into a recent VFM fine-tuning pipeline. Together, these analyses provide a more targeted view of both the limitations of existing foundation-model-based methods and the compatibility of UrbanGenNet with this paradigm.

1) GeoSA-BaSA Versus UrbanGenNet (Small-Object Completeness Under Shadow-Like Perturbations): We further compare UrbanGenNet with GeoSA-BaSA using fine-grained object- and small-structure-oriented metrics. In addition to the clean setting, we introduce two shadow-like perturbations, namely, *shadow* and *irregular-shadow*, to examine model behavior under spatially heterogeneous radiometric shifts. These perturbations are deliberately chosen because they directly probe local illumination disturbances, which are particularly relevant to high-resolution remote sensing imagery. The results are reported in Table III.

Under the clean setting, GeoSA-BaSA remains stronger in holistic benchmark-oriented metrics, achieving higher boundary IoU and object-level F1 (61.00 versus 58.62 and 72.70 versus 68.52, respectively). However, UrbanGenNet already shows a more favorable trend in small-object completeness, with slightly higher small-object recall (60.42 versus 60.33), lower miss rate (39.58 versus 39.67), and higher mean best IoU (43.36 versus 42.42). This indicates that, even without artificial perturbation, UrbanGenNet is less prone to missing weakly visible or small-scale building instances.

The robustness advantage of UrbanGenNet becomes more evident under shadow-like perturbations. As shown by the parenthesized changes in Table III, GeoSA-BaSA remains stronger in absolute boundary IoU and object-level F1, but its B-IoU and Obj-F1 slightly decrease under both perturbation settings. In contrast, UrbanGenNet remains stable or even slightly improves, with changes of +0.10/+0.48 under *shadow* and +0.04/+0.10 under *irregular-shadow* for B-IoU/Obj-F1, respectively. This suggests that UrbanGenNet does not surpass GeoSA-BaSA in absolute boundary- and object-level quality, but exhibits more stable local structural behavior under radiometric perturbations.

Moreover, the small-object-oriented metrics further support this complementary advantage. Under *shadow*, UrbanGenNet improves over GeoSA-BaSA by 1.29 points in small-object recall, reduces the miss rate by 1.29 points, and achieves a 1.77-point gain in small-object mean best IoU. A similar tendency is observed under *irregular-shadow*, where

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

TABLE III

FINE-GRAINED COMPARISON BETWEEN GEOSA-BASA AND URBANGENNET ON WHU-MIX UNDER CLEAN AND SHADOW-LIKE PERTURBATION SETTINGS. BEST RESULTS WITHIN EACH SETTING ARE HIGHLIGHTED IN BOLD

Setting	Method	B-IoU	Obj-F1	S-Recall	S-Miss↓	S-BestIoU
Clean	GeoSA-BaSA	**61.00**	**72.70**	60.33	39.67	42.42
	UrbanGenNet	58.62	68.52	**60.42**	**39.58**	**43.36**
Shadow	GeoSA-BaSA	**60.91** (-0.09)	**72.65** (-0.05)	59.73	40.27	42.11
	UrbanGenNet	58.72 (+0.10)	69.00 (+0.48)	**61.02**	**38.98**	**43.88**
Irreg.-Shadow	GeoSA-BaSA	**60.98** (-0.02)	**72.69** (-0.01)	60.13	39.87	42.30
	UrbanGenNet	58.66 (+0.04)	68.62 (+0.10)	**60.76**	**39.24**	**43.66**

B-IoU denotes Boundary IoU, S-Recall denotes small-object recall, S-Miss denotes small-object miss rate, and S-BestIoU denotes small-object mean best IoU. Parentheses in B-IoU and Obj-F1 under perturbation settings denote the change relative to the corresponding clean setting of the same method.

TABLE IV

HYBRID VALIDATION UNDER A UNIFIED DINOV2-BASED PROTOCOL ON WHU-MIX. BEST PERFORMANCE IS HIGHLIGHTED IN BOLD

Method	Trainable Params (M)	mIoU	US_Kitsap	CN_Wuxi	NZ_Dunedin	DE_Potsdam	SD_Khartoum
Rein	2.99	79.15	83.44	86.63	88.29	76.93	60.44
Rein + SASP	2.99	81.44	**85.60**	**87.10**	**89.38**	76.39	67.61
Rein + Full	2.99	**81.53**	85.04	86.55	89.25	76.29	**70.09**
GeoSA-BaSA	16.99	81.11	84.61	86.60	89.03	**79.31**	66.02

All methods in this table are evaluated under a DINOv2-based setting. Trainable parameter counts are reported for reference.

UrbanGenNet continues to outperform GeoSA-BaSA on all three small-object-oriented metrics.

These results suggest that the main advantage of UrbanGenNet does not lie in surpassing GeoSA-BaSA in holistic benchmark scores, but in preserving small-object completeness under spatially nonstationary radiometric shifts. While the standalone UrbanGenNet setting remains weaker than GeoSA-BaSA in absolute boundary IoU and Obj-F1, the parenthesized changes indicate more stable boundary- and object-level behavior under perturbations. To examine whether this local-structure benefit can be further strengthened in a stronger foundation-model fine-tuning pipeline, we provide an additional rein-based fine-grained verification in Section IV-D2.

To further illustrate the behavior discussed above, Fig. 6 presents representative comparisons on small buildings under both clean and shadow-perturbed conditions. In challenging local regions, UrbanGenNet generally preserves small building instances more completely than GeoSA-BaSA. This difference becomes more evident under shadow perturbation, where GeoSA-BaSA more often exhibits degraded local predictions, including missed structures, fragmented masks, and occasional spurious extra predictions around shadow-affected areas. By contrast, UrbanGenNet tends to maintain more compact and coherent predictions, yielding a better balance between small-object completeness and local precision.

2) Compatibility With Foundation-Model Fine-Tuning (Hybrid and Fine-Grained Validation on Rein): To examine whether the proposed design remains effective when coupled with a recent foundation-model-based fine-tuning pipeline, we further conduct a hybrid validation based on rein. Specifically, under a unified DINOv2-based protocol, we compare the reproduced rein baseline, rein + SASP, and rein + the full UrbanGenNet design. We additionally include GeoSA-BaSA as a strong remote sensing-specific reference. The holistic segmentation results are summarized in Table IV. To further examine whether the proposed components can improve local structural quality in this stronger foundation-model fine-tuning setting, we also report fine-grained object- and small-structure-oriented metrics for rein and rein + full in Table V.

It is worth noting that all rein-based variants use the same trainable parameter budget (2.99 M). Hence, the improvements of rein + SASP and rein + full cannot be explained by a larger number of trainable parameters, and are more plausibly associated with the proposed design. Under this matched parameter budget, both rein + SASP and rein + full consistently improve over the reproduced rein baseline, indicating that explicit spatial style perturbation and decoupled invariance learning remain beneficial even when coupled with a strong pretrained backbone.

Among the rein-based variants, rein + full achieves the best overall mIoU, while rein + SASP attains the strongest performance on several target domains, including US_Kitsap, CN_Wuxi, and NZ_Dunedin. The gain is particularly notable on the most challenging target domain, SD_Khartoum. Compared with the reproduced rein baseline, rein + full improves the IoU on SD_Khartoum from 60.44% to 70.09%, yielding a substantial gain of 9.65 points. It also surpasses the GeoSA-BaSA reference on this domain by 4.07 points. These results suggest that the proposed design is not limited to the original ConvNeXt-B setting, but can also serve as a complementary module on top of recent foundation-model-inspired fine-tuning pipelines, especially under severe spatially nonstationary radiometric shifts.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

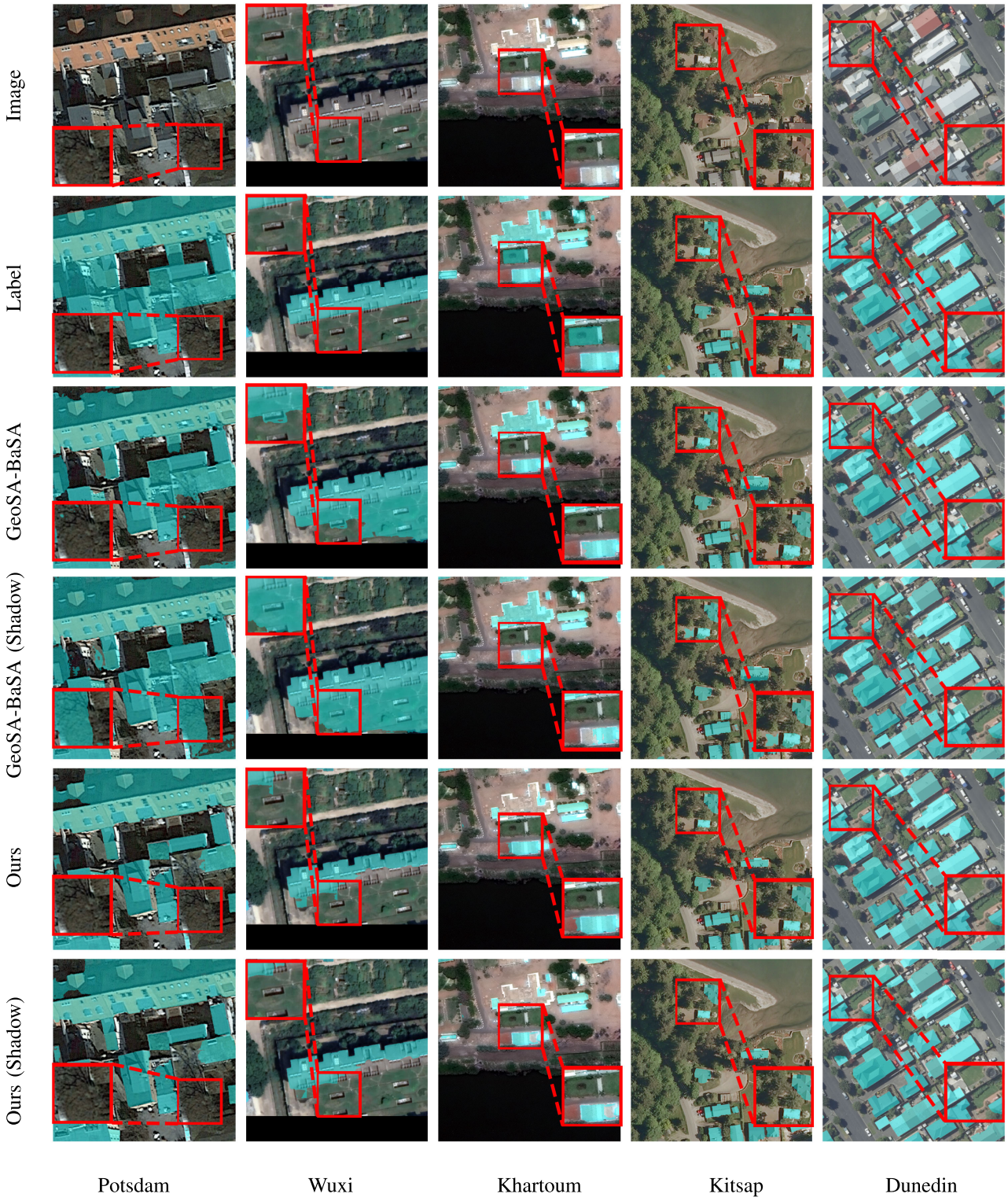


Fig. 6. Qualitative comparison between GeoSA-BaSA and UrbanGenNet on representative small-object cases from the five WHU-Mix target cities. The rows "GeoSA-BaSA (shadow)" and "Ours (shadow)" denote predictions under shadow perturbation. Red boxes highlight challenging local regions where differences in small-object completeness, local fragmentation, and weak-structure preservation are most visible.

Beyond the holistic mIoU comparison, we further evaluate the rein-based variants using the same fine-grained metrics as in Table III. As shown in Table V, rein + full consistently improves boundary IoU, Obj-F1, and S-BestIoU over rein under clean, shadow, and irregular-shadow settings. The gains are +3.79 to +3.97 points for boundary IoU, +2.57 to +2.75 points for Obj-F1, and +3.62 to +3.86 points for S-BestIoU. S-Recall and S-Miss are also slightly improved under the clean and shadow settings, while remaining comparable with only a marginal tradeoff under irregular-shadow.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

TABLE V

REIN-BASED FINE-GRAINED VERIFICATION ON WHU-MIX UNDER CLEAN AND SHADOW-LIKE PERTURBATIONS. BEST RESULTS WITHIN EACH SETTING ARE HIGHLIGHTED IN BOLD

Setting	Method	B-IoU	Obj-F1	S-Recall	S-Miss↓	S-BestIoU
Clean	Rein	58.03	77.37	61.05	38.95	39.73
	Rein+Full	**61.75** (+3.72)	**79.94** (+2.57)	**61.14** (+0.09)	**38.86** (-0.09)	**43.35** (+3.62)
Shadow	Rein	57.64	76.91	61.17	38.83	39.69
	Rein+Full	**61.61** (+3.97)	**79.66** (+2.75)	**61.21** (+0.04)	**38.79** (-0.04)	**43.55** (+3.86)
Irreg.-Shadow	Rein	57.91	77.18	**61.35**	**38.65**	39.88
	Rein+Full	**61.70** (+3.79)	**79.83** (+2.65)	61.32 (-0.03)	38.68 (+0.03)	**43.54** (+3.66)

Parenthesized values indicate the absolute change of Rein+Full relative to Rein under the same setting.

TABLE VI

ABLATION STUDY OF THE PROPOSED MODULES ON WHU-MIX. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	mIoU	US_Kitsap	CN_Wuxi	NZ_Dunedin	DE_Potsdam	SD_Khartoum
Baseline	71.07	82.59	85.49	89.87	52.87	41.17
+ Urban Loss	73.81	82.37	82.78	87.32	61.61	51.28
+ SASP	75.12	**83.92**	**85.96**	**89.91**	62.28	51.90
Full Model (coupled)	72.71	80.96	84.07	88.24	56.21	52.96
Full Model	**76.10**	83.35	85.21	89.57	**66.07**	**54.61**

It is also worth noting that rein + full achieves higher boundary IoU than GeoSA-BaSA under all three settings. Meanwhile, rein already attains higher Obj-F1 than GeoSA-BaSA, and rein + full further improves this metric. It shows that the proposed components can further strengthen boundary/object-level refinement and small-object localization quality when integrated into a stronger foundation-model fine-tuning pipeline.

E. Ablation Studies and Analysis

To validate the intrinsic mechanisms and design choices of UrbanGenNet, we conduct extensive ablation studies on WHU-Mix and FLAIR #1.

1) Effectiveness of Proposed Modules: Table VI reports the incremental contributions of the proposed modules on WHU-Mix. Starting from the baseline ERM model, introducing urban loss improves the mIoU from 71.07% to 73.81%, indicating that explicitly encouraging class-wise invariant structure learning is beneficial for cross-city generalization. Introducing SASP yields a larger gain, improving the mIoU to 75.12%, which suggests that modeling spatially nonstationary radiometric perturbations is particularly effective for this benchmark.

When both modules are combined, the full UrbanGenNet achieves the best overall performance, reaching 76.10% mIoU. The gains on NZ_Dunedin, US_Kitsap, and CN_Wuxi are relatively limited because these target domains are visually closer to the source distribution and the baseline already achieves relatively high performance on them, leaving less room for improvement than on more challenging domains. Importantly, the full model delivers the strongest results on the two more challenging target domains, DE_Potsdam and SD_Khartoum, where the IoU increases to 66.07% and 54.61%, respectively. By contrast, SASP alone already attains the strongest performance on several relatively easier domains, including US_Kitsap, CN_Wuxi, and NZ_Dunedin. This pattern suggests that SASP provides a strong improvement in handling appearance variation, while the additional urban loss further enhances robustness in more difficult cross-domain scenarios. Overall, the results confirm that the two modules are complementary rather than redundant. To further make the component-level behavior visually interpretable, Fig. 7 provides a visual ablation comparison on representative WHU-Mix target-domain samples.

A natural alternative to the proposed design is to directly impose the class-wise invariance constraint on the perturbed branch, i.e., to optimize appearance perturbation and invariance learning within a single coupled stream. We therefore include the full model (coupled) variant in Table VI, where urban loss is applied to the SASP-perturbed stream without stream decoupling.

The coupled variant achieves 72.71% mIoU, which is 3.39 points lower than the proposed decoupled full model (76.10%). It is also 2.41 points lower than the +SASP variant (72.71% versus 75.12%). This suggests that directly adding the class-wise invariance objective to the perturbed stream can partially offset the benefit of radiometric diversification, rather than producing the complementary gain obtained through stream decoupling. These results provide empirical support for computing urban loss on the clean stream while applying SASP to the augmented stream.

In order to examine the effect of stream coupling on optimization stability, Fig. 8 compares the coupled and decoupled training variants. As shown in Fig. 8(a), both variants converge, but the coupled variant maintains a higher postwarm-up loss and exhibits more pronounced intermittent spikes, whereas the decoupled variant follows a lower and smoother trajectory. Fig. 8(b) further quantifies this difference: the standard deviation of the detrended loss residual is 0.0430 for the coupled variant and 0.0133 for the decou-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

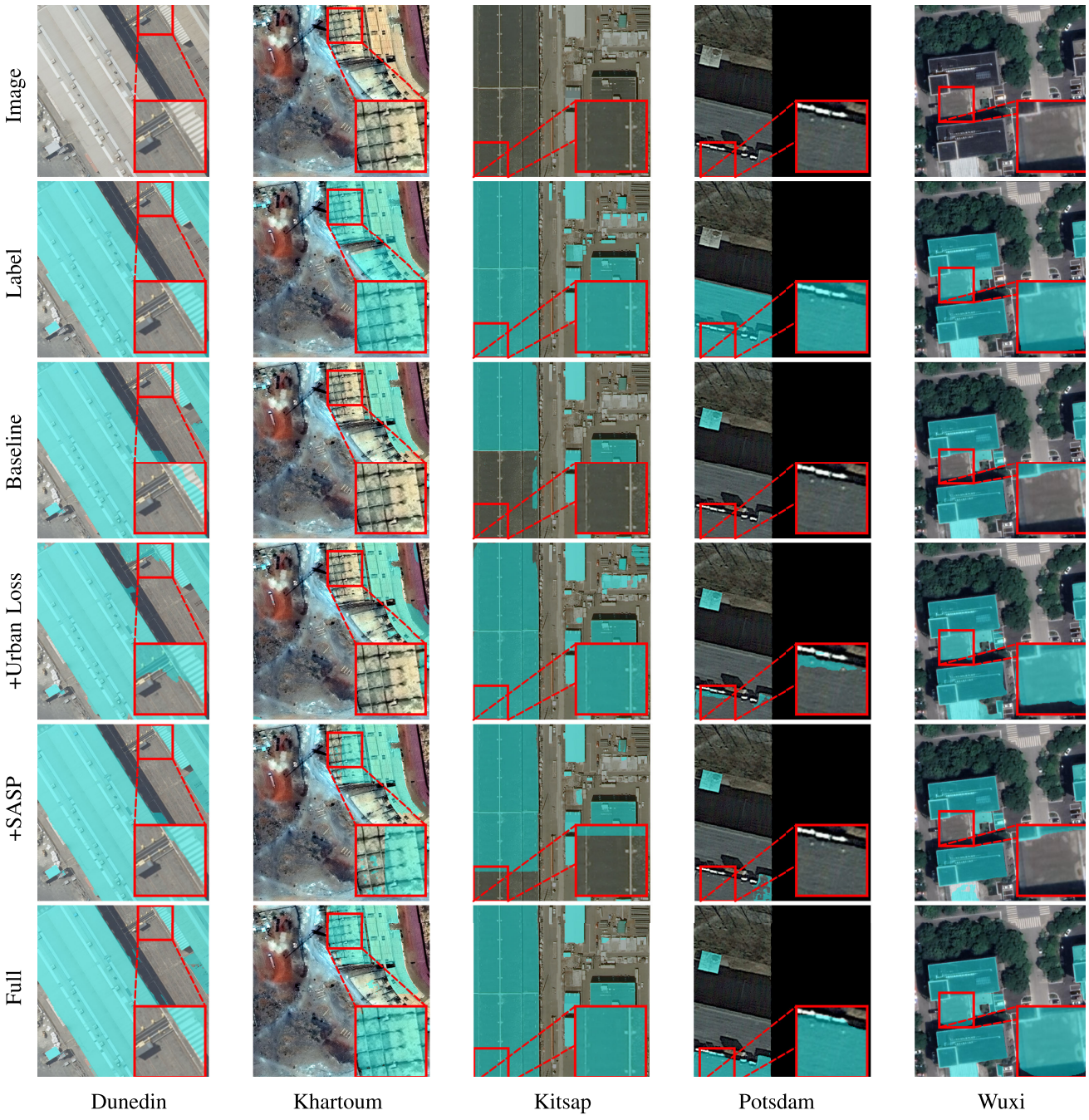


Fig. 7. Visual ablation comparison on WHU-Mix.

pled variant, while the corresponding 95th percentiles of the absolute residual are 0.0734 and 0.0260, respectively. Together with the accuracy comparison in Table VI, these results provide complementary empirical support for the proposed stream decoupling.

To further verify that this behavior is not limited to binary building extraction, we additionally report a compact module ablation on the FLAIR #1 geographic generalization split in Table VII. The same overall trend is observed: both urban loss and SASP contribute positively, and their combination yields the strongest overall performance. The most pronounced category-wise gains are observed on *bare soil*, *vineyard*, and *pervious surface*. In particular, *bare soil* improves substantially from 28.13% in the baseline to 42.12% in the full model, while *vineyard* and *pervious surface* also exhibit consistent gains. These results provide additional evidence that the proposed design generalizes beyond WHU-Mix to more complex multiclass land-cover mapping scenarios.

In addition to the segmentation accuracy, Table VII also reports the online training-time overhead and peak GPU memory on FLAIR #1. Urban loss introduces only a small additional cost, increasing the training overhead from 1.00× to 1.06× and the peak memory from 25.59 to 27.74 GB. SASP incurs a moderate training-time overhead of 1.42×,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

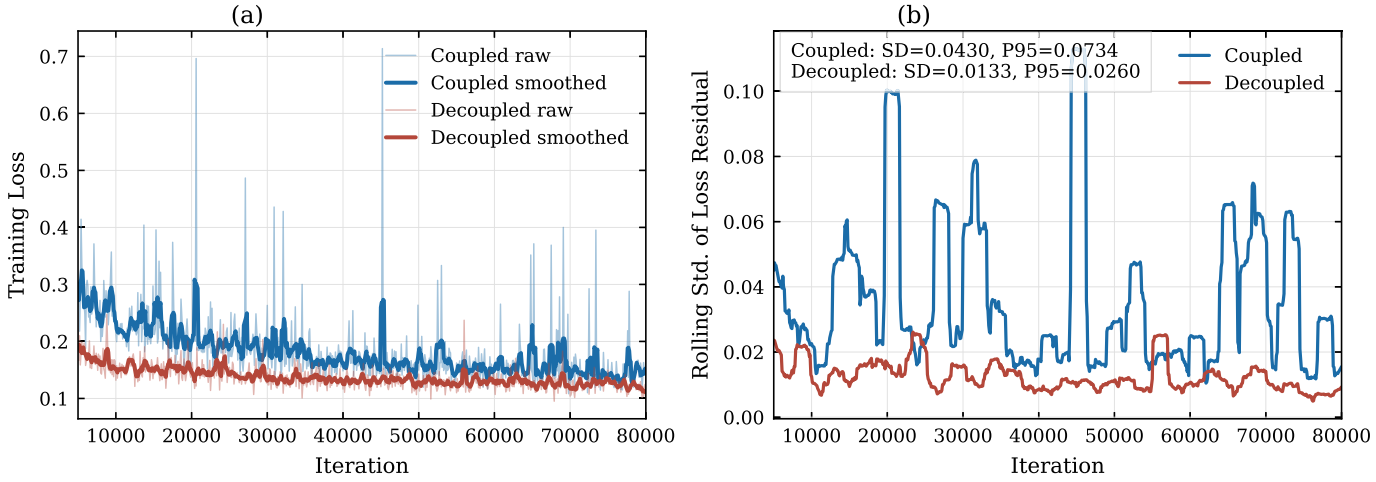


Fig. 8. Optimization-stability comparison between the coupled and decoupled formulations. (a) Postwarm-up training-loss trajectories, where the raw losses are shown at lower opacity and the smoothed curves indicate the overall trends. Although both formulations converge, the coupled formulation maintains a higher loss level and exhibits more pronounced intermittent spikes, whereas the decoupled formulation follows a lower and smoother trajectory. (b) Rolling standard deviation of the detrended loss residual. The coupled formulation shows consistently larger and more frequent fluctuations, quantitatively confirming the improved optimization stability of the decoupled formulation.

TABLE VII

MODULE ABLATION AND TRAINING-TIME COST ON THE FLAIR #1. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	mIoU	Bare Soil	Vineyard	Pervious Surface	Train Overhead	Memory Max (GB)
Baseline	53.78	28.13	70.30	48.59	1.00×	25.59
+ Urban Loss	54.17	31.77	71.21	48.32	1.06×	27.74
+ SASP	55.04	38.89	73.63	51.77	1.42×	27.73
Full Model	**55.97**	**42.12**	**75.94**	**52.34**	2.08×	56.01

while its peak memory remains comparable to the urban loss variant at 27.73 GB. The full UrbanGenNet requires 2.08× training overhead and 56.01-GB peak memory, mainly due to the joint clean/augmented training streams and the additional regularization during training. These results provide a more complete practical efficiency profile: although the proposed framework increases training-time cost, all additional components are training-only, and the inference-time pipeline remains unchanged.

2) Analysis of Latent Environment Discovery: Since urban loss relies on latent environment partitions, we further analyze the sensitivity and stability of the latent environment discovery module. To isolate the role of environment discovery from spatial style perturbation, we conduct this sensitivity analysis on the +Urban Loss variant without SASP. As shown in Fig. 9, the connected circular markers report the downstream mIoU under different numbers of latent environments K. The best performance is obtained at $K = 5$, with an mIoU of 73.81%. When K is smaller or larger, the performance decreases to 71.83%, 72.17%, 72.47%, and 71.66% for $K = 3, 4, 6, 7$, respectively.

These results indicate that the environment granularity affects the effectiveness of class-wise invariant regularization. Too few environments may merge heterogeneous urban contexts into the same group, while too many environments may over-fragment the source domains and reduce the reliability of class–environment bucket estimation. Therefore, we adopt $K = 5$ as the default setting in UrbanGenNet.

To examine the variance across random seeds, we further evaluate the downstream performance at $K = 5$ using different random seeds. The additional markers around $K = 5$ in Fig. 9 denote these extra seed runs. The mIoU values are 73.31%, 73.88%, and 73.81% for seeds 0, 2, and 42, respectively, yielding a mean of 73.67%, a standard deviation of 0.31, and a range of 0.57 points. This suggests that the downstream performance of the environment-based cIRM regularization remains stable across different random seeds and does not exhibit large seed-induced variance in this setting.

To better understand what the discovered environments represent, we further analyze the $K = 5$ latent partition from both quantitative and qualitative perspectives. Here, an "environment" does not denote a manually annotated city label or an end-to-end trainable parameter. Instead, it refers to a metadata-free latent partition obtained from CLIP-guided background/context embeddings.

Fig. 10 shows the domain–environment assignment distribution. The results indicate that the discovered environments are not simply equivalent to explicit domain labels. For example, several domains, such as Chongqing, Hangzhou, Tyrol, and Global, are distributed across multiple latent environments, while Env 0 and Env 2 contain samples from multiple domains. At the same time, some domains or environments remain dominated by particular assignments, which is expected because geographic domains naturally correlate with context and appearance in remote sensing imagery. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

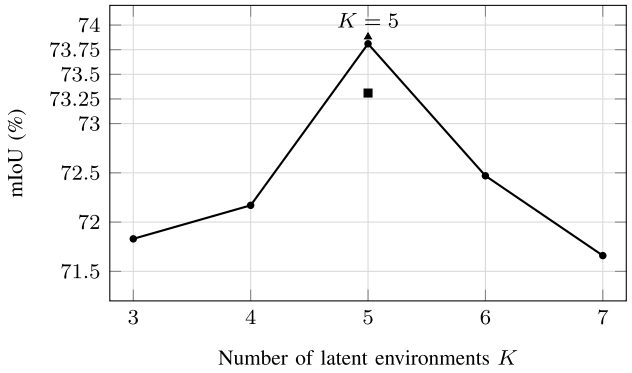


Fig. 9. Sensitivity and seed-stability analysis of latent environment discovery on WHU-Mix using the +Urban Loss variant. The additional markers at $K = 5$ denote extra random-seed runs.

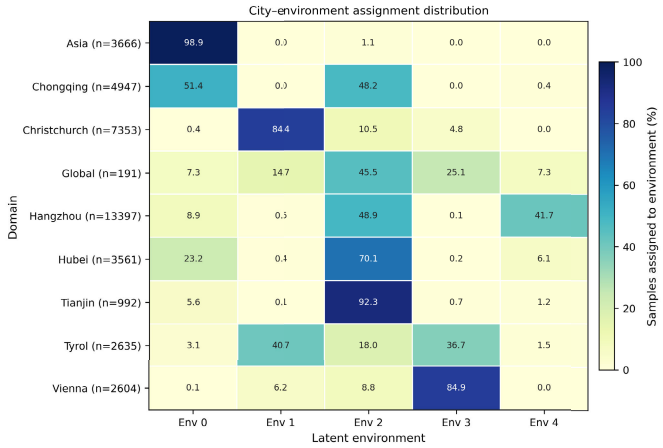


Fig. 10. Domain–environment assignment distribution for the $K = 5$ latent partition. Values denote the percentage of samples from each domain assigned to each latent environment. The results show that the discovered environments are not simply equivalent to explicit domain labels, but capture both domain-specific and cross-domain contextual variation.

suggests that the latent environments capture both domain-specific and cross-domain contextual variation.

Fig. 11 further presents representative cross-domain prototypes for the discovered latent environments. Several environments exhibit visually coherent contextual patterns across different domains. For example, Env 2 mainly contains bright industrial roofs, large impervious surfaces, and road-like structures from multiple domains, while Env 1 tends to group suburban or low-rise built-up contexts mixed with vegetation. These qualitative observations suggest that the discovered environments encode meaningful urban/contextual patterns for class-wise invariant regularization. We note that not all environments are equally cross-domain; some are more domain-dominated, reflecting the natural correlation between geography, scene layout, and appearance.

3) Feature Space Analysis: To better understand the feature-level effect of the proposed framework, we visualize the domain subcentroids using t-SNE and report the corresponding maximum mean discrepancy (MMD) scores in Fig. 12. While t-SNE mainly serves as a qualitative visualization tool, the accompanying MMD values provide a complementary quantitative measure of source–target discrepancy.

TABLE VIII

COMPATIBILITY OF URBANGENNET ACROSS DIFFERENT BACKBONE ARCHITECTURES ON WHU-MIX. "BASELINE" DENOTES THE CORRESPONDING ERM MODEL BUILT ON EACH BACKBONE, AND "GAIN" DENOTES THE IMPROVEMENT BROUGHT BY URBANGENNET OVER THE CORRESPONDING BASELINE. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Backbone	Baseline	UrbanGenNet	Gain
ConvNeXt	**71.07**	**76.10**	**+5.03**
ViT	67.44	70.25	+2.81
Vim	65.44	67.36	+1.92

Compared with the baseline, both urban loss and SASP reduce the discrepancy between source and target representations, as reflected by the monotonic decrease of the MMD score from 0.537 to 0.429 and 0.376, respectively. The full model further lowers the MMD value to 0.362 and exhibits the most mixed source–target distribution in the visualization. These results provide complementary evidence that UrbanGenNet improves cross-domain alignment in the learned representation space, with the strongest effect achieved when radiometric diversification and invariance learning are combined.

4) Compatibility Across Diverse Backbones: To examine whether the proposed framework depends on a specific backbone family, we replace the default ConvNeXt backbone with representative alternatives from transformer-based and state-space-based architectures, namely, ViT-B [36] and Vim-S [37]. The results are reported in Table VIII.

UrbanGenNet consistently improves out-of-domain performance across all three backbone types. In particular, the gain is strongest on ConvNeXt (+5.03 mIoU), while consistent improvements are also observed on ViT-B (+2.81) and Vim-S (+1.92). These results indicate that the proposed design is not restricted to a single architectural family, although its effectiveness is more pronounced on some backbones than on others.

A closer inspection suggests that the larger gain on ConvNeXt is consistent with the fact that UrbanGenNet is instantiated with ConvNeXt as its default backbone, whereas ViT-B and Vim-S serve here as representative non-CNN alternatives. Nevertheless, the positive gains on both transformer- and state-space-based backbones still demonstrate that the proposed SASP and urban loss are transferable beyond the original CNN setting. This cross-architecture consistency supports the view that UrbanGenNet captures a generally useful DG mechanism, rather than merely exploiting a specific backbone architecture.

5) Objective Function Analysis: We further analyze the composition of the proposed urban loss by decomposing it into its two key components, namely, the cIRM and the boundary-aware supervision term. Specifically, we compare four objective combinations: CE + Dice, CE + boundary IoU, cIRM + Dice, and urban loss (i.e., cIRM + boundary IoU). The results are reported in Table IX.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

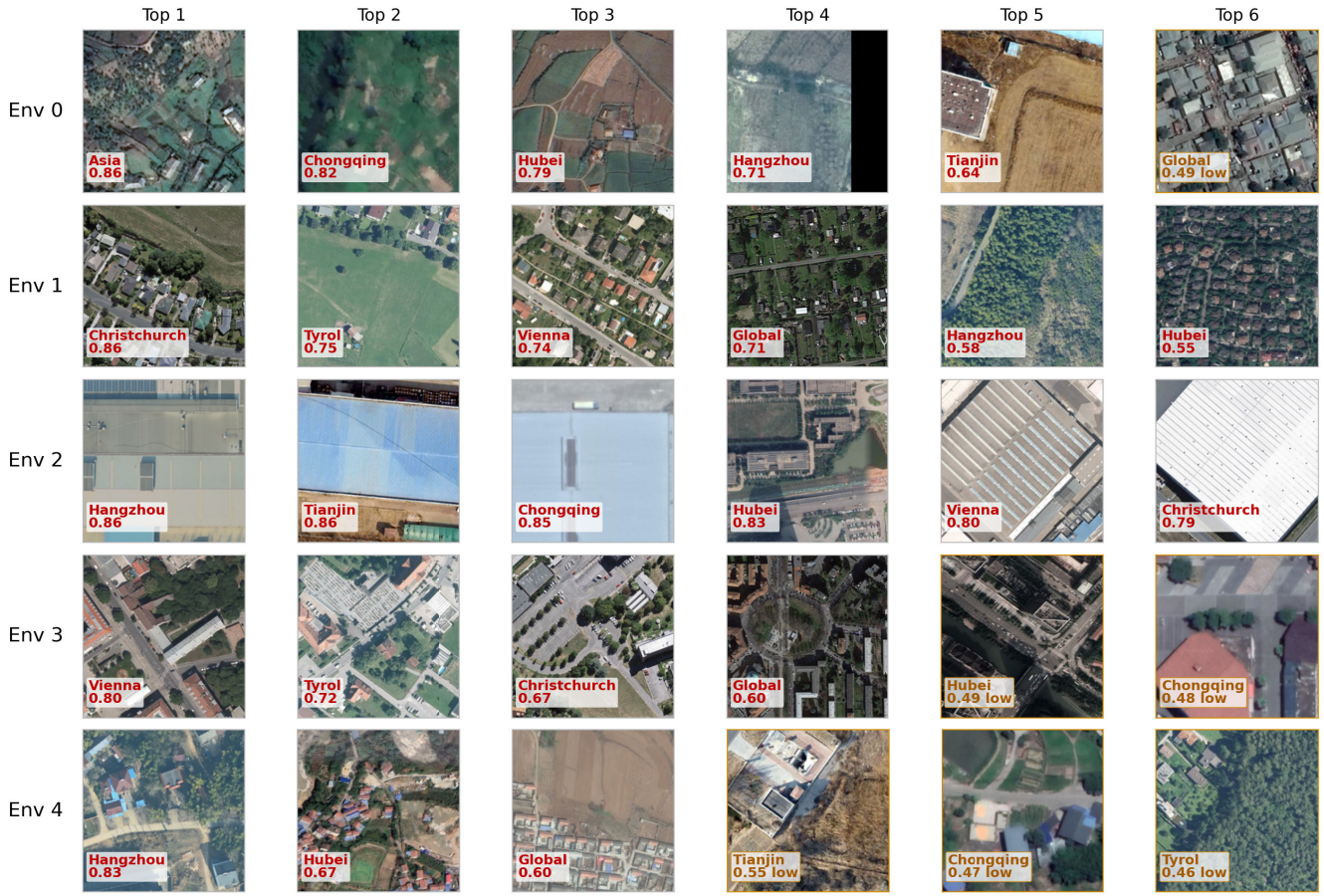


Fig. 11. Representative prototypes for the $K = 5$ latent environments. Each row corresponds to one latent environment. The bottom-left label in each patch denotes the source domain of the sample and its assignment score to the corresponding latent environment.

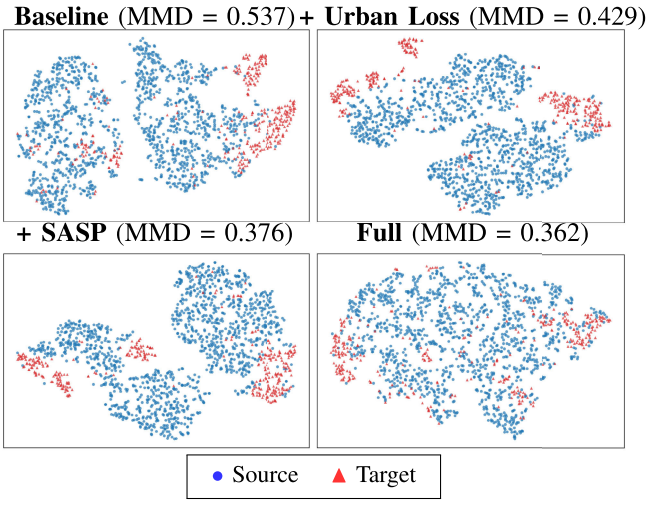


Fig. 12. t-SNE visualization of domain subcentroids for the baseline, +Urban Loss, +SASP, and the full model on WHU-Mix. Blue circles denote source samples and red triangles denote target samples. The corresponding MMD values are reported in each subfigure. Compared with the baseline, both urban loss and SASP reduce source–target discrepancy, while the full model yields the most mixed source–target distribution and the lowest MMD value.

Replacing Dice with boundary IoU yields consistent but moderate improvements over the corresponding CE-based baseline, indicating that boundary-aware supervision is beneficial for cross-domain performance. In contrast, introducing the cIRM leads to a larger gain in generalization. In particular, cIRM + Dice improves the mIoU from 71.85% to 73.40%, while further combining cIRM with boundary IoU yields the best overall result of 76.10%.

The advantage of the full objective is especially evident on the more challenging target domains, DE_Potsdam and SD_Khartoum, where the IoU reaches 66.07% and 54.61%, respectively. These results suggest that the two components of urban loss play complementary roles: the cIRM provides the major gain in generalization, while boundary-aware supervision further improves performance when combined with it. Overall, the best results are achieved when invariance

TABLE IX
ABLATION ON DIFFERENT OBJECTIVE COMBINATIONS ON WHU-MIX. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Loss	mIoU	DE_Potsdam	SD_Khartoum
CE + Dice	71.85	53.03	47.08
CE + Boundary IoU	72.42	55.11	47.26
cIRM + Dice	73.4	58.85	50.26
Urban Loss (ours)	**76.1**	**66.07**	**54.61**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

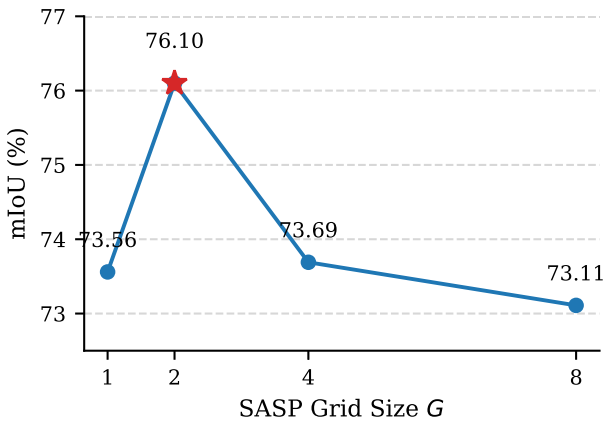


Fig. 13. Sensitivity analysis of SASP grid granularity on WHU-Mix. The model achieves the best performance at $G = 2$, indicating that a moderate spatial partition provides the most effective balance between perturbation diversity and semantic preservation.

TABLE X
SENSITIVITY ANALYSIS OF SASP HYPERPARAMETERS ON WHU-MIX. VALUES IN PARENTHESES DENOTE THE CHANGE RELATIVE TO THE DEFAULT SETTING

Hyperparameter	Value	mIoU
Noise intensity ϵ	0.0	73.37 (-1.75)
	0.1	73.80 (-1.32)
	0.3	**75.12**
	0.5	74.53 (-0.59)
Beta parameter α	0.1	74.09 (-1.03)
	0.5	**75.12**
	1.0	74.55 (-0.57)
Micro-noise grid k	8	73.51 (-1.61)
	16	**75.12**
	32	74.98 (-0.14)

learning and boundary-aware supervision are jointly incorporated into the final objective.

6) Sensitivity Analysis of SASP: The macro-level spatial trend in SASP is governed by the grid size $G \times G$, which determines the spatial granularity of the perturbation field. We therefore study the sensitivity to $G \in \{1, 2, 4, 8\}$ in Fig. 13. The results show a clear nonmonotonic trend. A moderate partition with $G = 2$ yields the best result of 76.10% mIoU, whereas both coarser and finer settings are less effective. Specifically, when $G = 1$, SASP collapses into a global style perturbation and cannot adequately capture spatially varying radiometric patterns, resulting in a lower mIoU of 73.56%. In contrast, when the grid becomes too fine ($G = 4$ and $G = 8$), the perturbation begins to rely on overly local statistics, which may absorb semantic structures and interfere with content preservation, leading to reduced performance of 73.69% and 73.11%, respectively. These observations suggest that $G = 2$ offers the most suitable balance between radiometric diversity and structural stability, which is consistent with the design motivation of SASP.

In addition to the grid size G, we further analyze three perturbation-related hyperparameters in SASP, including the micro-noise intensity ϵ, the Beta distribution parameter α for style interpolation, and the micro-noise grid size k. We vary one hyperparameter at a time while keeping the others fixed at the default setting. As shown in Table X, the default setting $\epsilon = 0.3$, $\alpha = 0.5$, and $k = 16$ achieves the best mIoU among the tested values. Removing the micro-noise field leads to a clear performance drop from 75.12% to 73.37%, suggesting that local low-frequency perturbations are useful for modeling spatially nonstationary radiometric variation. A smaller noise intensity $\epsilon = 0.1$ provides insufficient perturbation diversity, while a larger value $\epsilon = 0.5$ slightly degrades the performance, indicating that overly strong perturbations may disturb semantic structures.

For the Beta distribution parameter, both $\alpha = 0.1$ and $\alpha = 1.0$ perform worse than the default $\alpha = 0.5$, suggesting that either overly extreme or overly averaged style interpolation is less favorable. For the micro-noise grid size, $k = 16$ performs best, while $k = 32$ achieves a close result and $k = 8$ degrades more clearly. This indicates that SASP benefits from a moderate low-frequency noise scale: overly coarse perturbations may lack sufficient local variation, whereas an appropriate spatial scale better simulates spatially varying radiometric effects. Overall, these results show that SASP is not dependent on a fragile hyperparameter choice; instead, the default setting provides a stable balance between radiometric diversity and semantic-structure preservation.

V. CONCLUSION

In this article, we presented UrbanGenNet, a domain-generalized remote sensing semantic segmentation framework designed to address the coupled challenge of spatially nonstationary radiometric variation and geographic structural shift. The proposed method combines a SASP, a task-specific urban loss, and a decoupled two-stream training strategy to jointly enhance radiometric robustness and structure-oriented invariance learning.

Extensive experiments on WHU-Mix and FLAIR #1 demonstrate that UrbanGenNet achieves strong and consistent cross-domain performance across both building extraction and multiclass land-cover segmentation. Beyond holistic benchmark results, the proposed framework shows complementary benefits in small-object completeness and robustness under shadow-like perturbations, while the comparison with GeoSA-BaSA also indicates that boundary- and object-level quality remain areas for further improvement. Ablation and sensitivity analyses further support the effectiveness of the decoupled design and the remote sensing-specific motivation of SASP and urban loss.

In future work, we note that strong foundation-model fine-tuning baselines may still remain stronger on some holistic benchmark-oriented metrics, indicating a remaining gap and a promising direction for future integration.

REFERENCES

[1] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.

[2] M. Luo, S. Ji, and S. Wei, "A diverse large-scale building dataset and a novel plug-and-play domain generalization method for building extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4122–4138, 2023.

[3] K. Saenko, B. Kulis, M. Fritz, and T. Darrell, "Adapting visual category models to new domains," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2010, pp. 213–226.

[4] J. Hoffman et al., "CyCADA: Cycle-consistent adversarial domain adaptation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1989–1998.

[5] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 1180–1189.

[6] T. Wang, Z. Zhang, and K.-L. Tsui, "A deep generative approach for rail foreign object detections via semisupervised learning," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 459–468, Jan. 2023.

[7] T. Wang, Z. Zhang, and M. Xie, "An anomaly-free representation learning approach for efficient railway foreign object detection," *IEEE Trans. Ind. Informat.*, vol. 20, no. 10, pp. 12125–12135, Oct. 2024.

[8] Y. Na, J. H. Kim, K. Lee, J. Park, J. Y. Hwang, and J. P. Choi, "Domain adaptive transfer attack-based segmentation networks for building extraction from aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5171–5182, Jun. 2021.

[9] J. Chen et al., "Memory-contrastive unsupervised domain adaptation for building extraction of high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605615.

[10] L. Hoyer, D. Dai, and L. Van Gool, "DAFormer: Improving network architectures and training strategies for domain-adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9924–9935.

[11] L. Hoyer, D. Dai, and L. Van Gool, "HRDA: Context-aware high-resolution domain-adaptive semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, Jun. 2022, pp. 372–391.

[12] D. Zhao, S. Wang, Q. Zang, D. Quan, X. Ye, and L. Jiao, "Towards better stability and adaptability: Improve online self-training for model adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11733–11743.

[13] D. Zhao et al., "Learning pseudo-relations for cross-domain semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 19134–19146.

[14] J. Fu, Y. Zhong, and F. Yang, "Adversarial domain generalization with MixStyle," in *Proc. Int. Conf. Adv. Robot. Mechatronics (ICARM)*, Jul. 2022, pp. 379–385.

[15] S. Lee, H. Seong, S. Lee, and E. Kim, "WildNet: Learning domain generalized semantic segmentation from the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9936–9946.

[16] X. Pan, P. Luo, J. Shi, and X. Tang, "Two at once: Enhancing learning and generalization capacities via IBN-Net," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 464–479.

[17] S. Choi, S. Jung, H. Yun, J. T. Kim, S. Kim, and J. Choo, "RobustNet: Improving domain generalization in urban-scene segmentation via instance selective whitening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11575–11585.

[18] D. Peng, Y. Lei, M. Hayat, Y. Guo, and W. Li, "Semantic-aware domain generalized segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2594–2605.

[19] D. Li, Y. Yang, Y.-Z. Song, and T. M. Hospedales, "Learning to generalize: Meta-learning for domain generalization," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2018, vol. 32, no. 1, pp. 3490–3497.

[20] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," 2019, *arXiv:1907.02893*.

[21] Z. Wei et al., "Stronger, fewer, & superior: Harnessing vision foundation models for domain generalized semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28619–28630.

[22] Y. Benigmim, S. Roy, S. Essid, V. Kalogeiton, and S. Lathuilière, "Collaborating foundation models for domain generalized semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3108–3119.

[23] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748–8763.

[24] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[25] R. Iizuka, J. Xia, and N. Yokoya, "Frequency-based optimal style mix for domain generalization in semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4501114.

[26] E. Ghazaei and E. Aptoula, "Multi-layer domain generalization for the semantic segmentation of optical remote sensing images," in *Proc. Joint Urban Remote Sens. Event (JURSE)*, May 2025, pp. 1–4.

[27] M. Luo, Y. Zan, K. Khoshelham, and S. Ji, "Domain generalization for semantic segmentation of remote sensing images via vision foundation model fine-tuning," *ISPRS J. Photogramm. Remote Sens.*, vol. 230, pp. 126–146, Dec. 2025.

[28] M. Oquab et al., "Dinov2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[29] Z. Gong et al., "CrossEarth: Geospatial vision foundation model for domain generalizable remote sensing semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 5, pp. 5147–5164, May 2026.

[30] A. Garioud, S. Peillet, E. Bookjans, S. Giordano, and B. Wattrelos, "Flair# 1: Semantic segmentation and domain adaptation dataset," 2022, *arXiv:2211.12979*.

[31] B. Cheng, R. Girshick, P. Dollár, A. C. Berg, and A. Kirillov, "Boundary IoU: Improving object-centric image segmentation evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15334–15342.

[32] T. Matsuura and T. Harada, "Domain generalization using a mixture of multiple latent domains," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 11749–11756.

[33] H. Li, Y. F. Wang, R. Wan, S. Wang, T. Q. Li, and A. C. Kot, "Domain generalization for medical imaging classification with linear-dependency regularization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 3118–3129.

[34] Z. Wu, X. Wu, X. Zhang, L. Ju, and S. Wang, "SiamDoGe: Domain generalizable semantic segmentation using Siamese network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 603–620.

[35] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.

[36] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.

[37] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jul. 2024, pp. 62429–62442.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:09:53 UTC from IEEE Xplore. Restrictions apply.